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CSE 4102

PROJECT PROPOSAL REPORT

Automated Indoor Plant Care System

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1. PROBLEM STATEMENT

Indoor plants in homes, offices, and apartment balconies are increasingly popular in cities such as Dhaka, yet keeping them healthy remains difficult for busy occupants. Most indoor plants are watered manually on a fixed routine or by guesswork. Overwatering leads to waterlogged soil and root rot, while underwatering causes wilting, leaf drop, and eventually the death of the plant. When the owner is away for work, travel, or holidays, the plant may go without water for days.

Watering is only part of the problem. Indoor plants depend on suitable temperature, humidity, and light, and these conditions are hard to judge by eye inside a room. Extreme heat increases water loss, dry air stresses leaves, and poorly lit corners slow growth. Simple timer-based waterers apply water on a schedule regardless of the actual condition of the soil, and they usually do not check the water reservoir. If the reservoir runs empty, the pump keeps running dry and can burn out, and the owner is not told that the tank needs refilling.

Therefore, there is a need for a low-cost embedded system that continuously measures soil moisture, waters the plant only when it is needed, monitors the surrounding climate and the reservoir, and keeps the owner informed from anywhere. This project proposes an Automated Indoor Plant Care System using an ESP32 microcontroller, a capacitive soil moisture sensor, a DHT22 temperature and humidity sensor, an LDR light sensor, an ultrasonic water level sensor, and a relay-controlled water pump, together with a mobile dashboard, providing an affordable and real-time plant care solution for homes and offices.

2. OBJECTIVES

- To automatically water the plant when soil moisture drops below a preset threshold.
- To monitor temperature, humidity, and light levels for optimal indoor plant growth.
- To monitor the reservoir water level, prevent pump dry-run and burnout, and indicate when refilling is needed.
- To display real-time sensor data on a mobile dashboard and allow remote manual watering from a mobile device.
- To send instant alerts for critical dehydration, extreme temperature, and low tank water level.
- To develop a low-cost and expandable indoor plant care prototype.

3. METHODOLOGY

The development of the system will begin with a requirement-analysis and planning phase in which the required irrigation, climate-monitoring, reservoir-protection, and remote-access functionalities are identified, and the most suitable sensors and supporting components are selected based on cost, availability, and accuracy. An ESP32 is chosen as the central controller instead of an Arduino UNO because its built-in Wi-Fi is needed for the mobile dashboard and alerts, and a capacitive soil moisture sensor is preferred over a resistive probe because it does not corrode in wet soil. The overall system will then be designed by mapping out how the ESP32, the capacitive soil moisture sensor, the DHT22 sensor, the LDR light sensor, the HC-SR04 ultrasonic sensor, a 16x2 I2C LCD, a buzzer, three status LEDs (green for normal, blue for watering, red for alerts), a 1-channel relay module, and a mini DC submersible pump with tubing and a drip nozzle will be interconnected around a single indoor plant pot, along with the logic that will govern threshold comparisons, pump control, and alerts.

Once the design is finalized, the hardware will be physically assembled by connecting the sensors, LCD, indicators, and relay-driven pump to the ESP32 using a breadboard, jumper wires, and resistors. The soil moisture sensor and LDR will be connected to ADC1 pins of the ESP32, since the ADC2 pins cannot be used while Wi-Fi is active, and a resistor voltage divider will reduce the 5V echo signal of the ultrasonic sensor to a safe 3.3V level. The pump will draw its current from the 5V adapter through the relay rather than from the ESP32 pins. The ESP32 will then be programmed in Arduino C/C++ through the Arduino IDE with the ESP32 board package. The firmware will be organized as five independent tasks using FreeRTOS, the real-time operating system built into the ESP32 core. Each task reads its own sensor, checks its own condition, and responds in its own loop at its own rate (the DHT22, for example, can only be read about once every two seconds), so a slow sensor or a long watering cycle never delays any other check. The soil readings are converted into a moisture percentage using calibration values recorded in dry air and in water, and the ultrasonic distance is converted into a tank level percentage. The tasks share only a small set of data: the latest sensor readings, a pump-lock flag, and a manual-watering flag.

Each task handles one condition independently. The irrigation task sends a dehydration alert whenever the soil moisture falls below a critical level (initially 20%). If the moisture is below the lower threshold (initially 35%), or a manual watering request has been received, it switches the pump ON through the relay, lights the blue LED, and shows WATERING on the LCD, unless the pump-lock flag is set. The pump runs until the moisture reaches the upper threshold (initially 60%) or a maximum run time elapses. The gap between the two thresholds prevents the pump from switching on and off rapidly, and the maximum run time protects the plant from flooding if a sensor

fails. The tank task sets the pump-lock flag whenever the reservoir is below the minimum level and clears it once the tank is refilled, so the pump can never run dry; while the tank is low, it also switches on the red LED and buzzer, shows REFILL TANK on the LCD, and sends a low-tank alert. The climate task switches on the red LED and buzzer and sends a temperature alert whenever the temperature leaves the safe range (initially 15°C to 35°C). The light task records the light level, and the display task pushes all readings to the Blynk dashboard, updates the LCD and status LEDs, and passes manual watering requests from the mobile device to the irrigation task. Because no condition waits for another, several alerts can be raised at the same moment (for example, a low tank and a high temperature), and automatic watering continues locally even if the Wi-Fi connection is lost. All threshold values will be finalized during calibration and testing.

Figure 1 illustrates this operational workflow: after initialization, the system starts five tasks that run in parallel, and each task continuously reads its own sensor, checks its own condition, triggers the appropriate pump, visual, audible, or mobile response, and repeats independently of the others. Finally, the completed prototype will be tested and evaluated under a range of conditions, including dry and wet soil, an empty reservoir, heated air near the DHT22, a covered LDR, several hazards occurring at the same time, and a disconnected Wi-Fi network, to confirm that watering, alerts, and dashboard updates trigger correctly and that the system operates reliably.

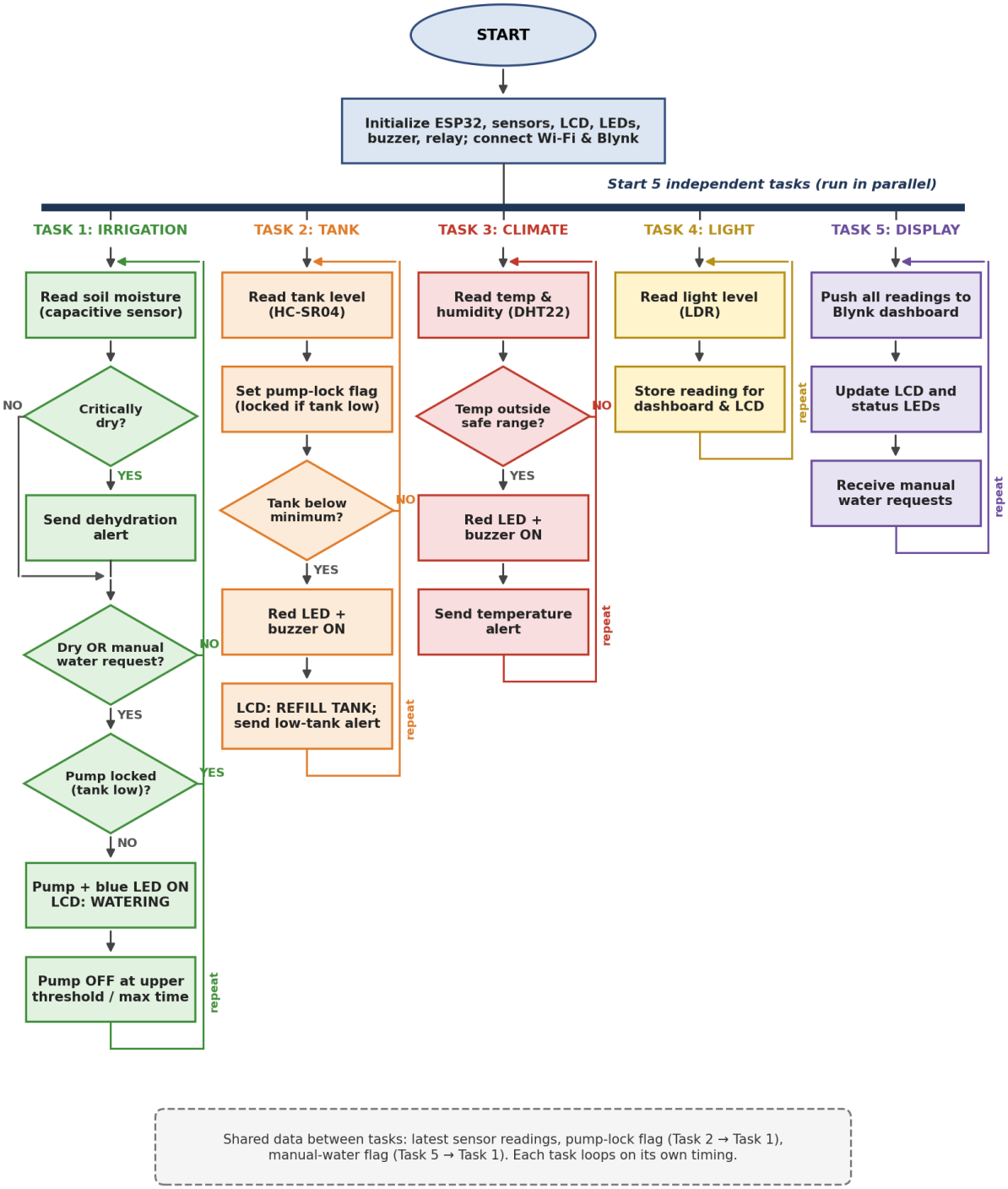


Figure 1: Parallel task workflow of the proposed Automated Indoor Plant Care System

4. HARDWARE AND SOFTWARE REQUIREMENTS

Component / Software	Qty.	Purpose
ESP32 DevKit V1 (Wi-Fi enabled)	1	Central controller; connects the system to the mobile dashboard
Capacitive Soil Moisture Sensor v1.2	1	Measures soil moisture in the plant pot
DHT22 Temperature & Humidity Sensor	1	Monitors air temperature and humidity around the plant
LDR Light Sensor Module	1	Measures the light level reaching the plant
HC-SR04 Ultrasonic Sensor	1	Measures the water level in the reservoir
1-Channel 5V Relay Module	1	Switches the water pump ON and OFF
Mini DC Submersible Water Pump (3-6V)	1	Pumps water from the reservoir to the plant
Silicone Tubing + Drip Nozzle	1 set	Delivers water to the soil around the plant
16×2 LCD with I2C	1	Displays sensor readings and system status locally
Buzzer	1	Provides an audible warning during critical conditions
LEDs with Resistors (Green, Blue, Red)	3 sets	Indicate normal, watering, and alert states
Resistors (1kΩ, 2kΩ)	1 set	Voltage divider for the ultrasonic sensor echo pin
Breadboard/PCB + Jumper Wires	1 set	Used for circuit assembly and connections
5V 2A Regulated Power Adapter	1	Powers the ESP32, sensors, and pump (via the relay)
Water Container (Reservoir) + Plant Pot	1 set	Stores water and holds the test plant
Arduino IDE + ESP32 Board Package	-	Used to write, compile, and upload the firmware (includes FreeRTOS)
C/C++	-	Programming language used for system development
Blynk IoT App + Blynk Library	-	Mobile dashboard, remote watering, and push notifications
DHT Sensor Library	-	Reads temperature and humidity from the DHT22
LiquidCrystal_I2C + Wire Library	-	Controls the LCD over I2C communication
Wokwi / Fritzing (Optional)	-	Used for circuit design, simulation, and documentation

Table 1: Hardware and Software Requirements

5. EXPECTED OUTCOMES

- Accurate measurement of soil moisture using the capacitive soil moisture sensor.
- Automatic watering when soil moisture drops below the preset threshold, stopping once the soil is sufficiently moist.
- Continuous monitoring of temperature, humidity, and light levels around the plant.
- Automatic pump lockout when the reservoir is low, with a clear refill indication on the LCD, LED, and mobile device.
- Real-time display of all sensor readings on the LCD and the mobile dashboard.
- Successful remote manual watering from a mobile device.
- Instant mobile alerts for critical dehydration, extreme temperature, and low tank water level.
- A functional and low-cost prototype for automated indoor plant care, monitoring, and remote control.

6. PROJECT SCHEDULE

The Gantt chart below illustrates the total project plan and scheduling on a week-by-week basis.

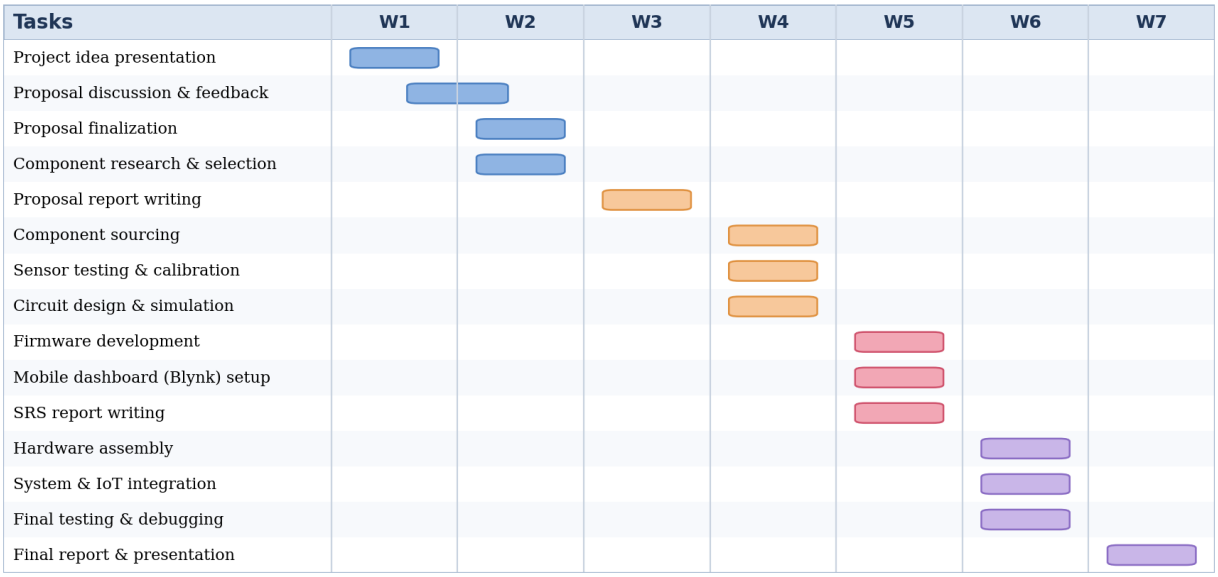


Figure 2: Gantt Chart for Automated Indoor Plant Care System

7. CONCLUSION

This proposal presents a low-cost embedded and IoT-based project built on the ESP32 platform, developed to automate the care of indoor plants in homes, offices, and other indoor spaces.

The Automated Indoor Plant Care System is designed to continuously monitor the plant's environment using a capacitive soil moisture sensor, a DHT22 temperature and humidity sensor, an LDR light sensor, and an ultrasonic water level sensor. When the soil becomes dry, the system will automatically water the plant through a relay-controlled pump and stop once the soil is sufficiently moist. The reservoir level will be checked before every watering cycle to protect the pump from running dry, and the owner will be notified when refilling is needed. Real-time readings will be shown on an LCD and a mobile dashboard, which will also allow the owner to water the plant remotely and receive instant alerts for critical dehydration, extreme temperature, and low tank water level.

The proposed system combines sensor-based monitoring, parallel task-based processing on the ESP32, automated irrigation, and IoT connectivity. A functional hardware prototype will be developed and tested under different soil, climate, and reservoir conditions to evaluate its sensing accuracy, watering response, alert reliability, and overall effectiveness.